Finite semidefinite representation of the simplest quantum correlation set

Abstract

The simplest Bell experiment allows two binary measurements on each of two separated systems. We prove that its complete quantum correlation set, including all marginal expectations and joint correlators, admits a finite semidefinite representation. Consequently, one fixed semidefinite program suffices to optimize every Bell functional in this scenario exactly. The proof reduces arbitrary-dimensional quantum realizations to real two-qubit correlations and constructs exact sums of squares through the singular limits in which measurements become compatible. The inequality coefficients and measurement parameters vary independently throughout the argument. A real-algebraic criterion converts these certificates into a finite global representation, whose matrices can be chosen with real algebraic entries.

1 Introduction

Which joint outcome probabilities can two separated quantum systems produce? Even when each party chooses between two binary measurements, the answer depends on both the correlations between the outcomes and their individual biases. Together these quantities form an eight-dimensional convex set. Its supporting hyperplanes give all quantum bounds on Bell inequalities in this experiment.

Semidefinite programming provides a systematic way to bound these quantum values. The Navascués–Pironio–Acín hierarchy imposes positivity on successively larger moment matrices [1]. In the two-input, two-output scenario, a different finite description follows from the reduction to two-qubit experiments [2]: the quantum set is the convex hull of a compact polynomial image. Its extreme points also admit an analytical characterization [3]. These descriptions make the set accessible to real algebraic geometry. They do not, by themselves, provide a finite linear matrix inequality for membership. Indeed, compact convex semialgebraic sets can fail to admit any semidefinite lift [4].

We prove that the complete quantum set in this scenario does admit such a lift. There is one finite positive-semidefinite constraint, with auxiliary real variables, whose projection is exactly the set of all eight expectations. The same feasible region therefore gives the exact quantum value of every Bell functional. The matrices can be chosen with real algebraic entries.

The difficulty occurs near compatible measurements. An inequality may approach a singular supporting hyperplane while an independent quantum experiment approaches a product-state or commuting-measurement boundary. A certificate valid at a fixed interior point need not remain valid through these simultaneous limits. We keep the coefficients of the inequality and the parameters of the experiment in separate real closed fields. Exact sums of squares in their tensor product then express the positivity needed for a finite lift. Scheiderer's tensor-evaluation criterion [5] turns this universal positivity statement into semidefinite representability.

We develop the physical reduction, the tensor criterion and the global assembly below. The local proof has two parts: regular contacts are controlled by curvature, whereas the compatible-measurement limits require explicit rescalings of the complete state polynomial. The accompanying *Complete proofs* supplies every local calculation and the source partitions used in the assembly.

2 The quantum set and the main theorem

Alice chooses $x \in \{0,1\}$ and Bob chooses $y \in \{0,1\}$. Their outcomes a, b take values in $\{+1,-1\}$. A quantum behavior has the form

$$p(a,b \mid x,y) = \mathrm{tr}\big[\rho(M_{a|x} \otimes N_{b|y})\big], \tag{1}$$

where ρ is a bipartite density operator and each measurement is a binary positive-operator-valued measurement. Arbitrary finite local Hilbert-space dimensions are allowed. Define the observables $A_x = M_{+|x} - M_{-|x}$ and $B_y = N_{+|y} - N_{-|y}$. Nonsignalling makes the four marginal expectations independent of the other party's input. We use the eight coordinates

$$q = (a_0, a_1, b_0, b_1, E_{00}, E_{01}, E_{10}, E_{11}), \qquad a_x = \langle A_x \rangle, \quad b_y = \langle B_y \rangle, \quad E_{xy} = \langle A_x B_y \rangle. \tag{2}$$

They recover every probability through

$$p(a,b \mid x,y) = \frac{1 + a a_x + b b_y + ab E_{xy}}{4}. \tag{3}$$

Let $\mathcal{Q}_{222} \subset \mathbb{R}^8$ denote the set obtained from Eq. (1).

Theorem 1 (Finite semidefinite representation)**.** *There exist finite integers N, p and real symmetric matrices $L_0, L_1, \ldots, L_8, M_1, \ldots, M_p$ of order N such that*

$$\mathcal{Q}_{222} = \left\{ q \in \mathbb{R}^8 : \ \exists y \in \mathbb{R}^p, \quad L_0 + \sum_{\nu=1}^{8} q_\nu L_\nu + \sum_{j=1}^{p} y_j M_j \succeq 0 \right\}. \tag{4}$$

All entries of these matrices may be chosen to be real algebraic numbers.

Thus $\mathcal{Q}_{222}$ is a spectrahedral shadow. For any Bell functional $\beta(q) = \beta_0 + \sum_\nu \beta_\nu q_\nu$, its maximum over Eq. (4) is exactly its quantum maximum. Compactness of $\mathcal{Q}_{222}$, proved next, ensures that this maximum is attained. The theorem establishes existence of a finite representation; evaluating its matrices and finding a useful matrix size are subsequent constructive questions.

3 A finite polynomial description of all quantum behaviors

The field-extension argument requires a fixed finite description of $\mathcal{Q}_{222}$. We obtain one by reducing the measurements before taking the convex hull.

For an effect $0 \preceq E \preceq I$, the matrix

$$\widehat{P}_E = \begin{pmatrix} E & \sqrt{E(I-E)} \\ \sqrt{E(I-E)} & I - E \end{pmatrix}, \qquad Vh = (h, 0), \tag{5}$$

is a projection and satisfies $V^* \widehat{P}_E V = E$. The embedding V is the same for both settings of a party. Applying it on both sides preserves all eight expectations. We may therefore use observables whose squares are the identity.

Two such observables have a common decomposition into invariant blocks of dimension at most two. To see this, let U, V be Hermitian involutions. The unitary $C = UV$ satisfies $UCU = C^*$. An eigenvector of C with nonreal eigenvalue λ pairs with its image under U, which has eigenvalue $\overline{\lambda}$; their span is invariant under both observables. On the $\lambda = \pm 1$ eigenspaces, $V = \pm U$, and one-dimensional blocks suffice. Repeating this decomposition gives the claimed blocks.

Within a two-dimensional block a choice of basis and phase puts the measurements in the form

$$A_0 = B_0 = Z, \qquad A_1 = c_A Z + s_A X, \qquad B_1 = c_B Z + s_B X, \quad c_A^2 + s_A^2 = c_B^2 + s_B^2 = 1, \quad (6)$$

where X, Z are the Pauli matrices. One-dimensional blocks embed in this description at $s_A = 0$ or $s_B = 0$. Since the measurements are block diagonal, their expectations are convex combinations of two-qubit expectations. The matrices in Eq. (6) are real. For each real symmetric measurement matrix M,

$$(r + is)^* M (r + is) = r^T M r + s^T M s. \quad (7)$$

Normalizing the nonzero real and imaginary parts separately therefore reduces complex states to mixtures of real pure states.

Let $\Phi : S^1 \times S^1 \times S^3 \to \mathbb{R}^8$ be the expectation map defined by Eq. (6) and a real unit state $\psi \in \mathbb{R}^4$. Every coordinate of Φ is polynomial. The preceding reduction and the physical realization of a finite mixture by direct sums give

$$\mathcal{Q}_{222} = \operatorname{conv} \Phi(S^1 \times S^1 \times S^3) = \left\{ \sum_{i=1}^{9} w_i \Phi(z_i) : w_i \geq 0,\ \sum_i w_i = 1,\ z_i \in S^1 \times S^1 \times S^3 \right\}. \quad (8)$$

The bound of nine is the Carathéodory bound in eight dimensions. It limits the number of atoms in a representation; the states and measurement angles still range continuously.

Equation (8) is a polynomial formula with rational coefficients and finitely many variables. Its domain is compact over $\mathbb{R}$, so $\mathcal{Q}_{222}$ is compact. Projection of a semialgebraic set is semialgebraic [6]. We can consequently transfer this same finite formula to any real closed extension K of $\mathbb{R}$; the resulting set is denoted $(\mathcal{Q}_{222})_K$. This supplies the precise meaning of a quantum point over K in the next section.

4 From Bell bounds to a finite lift

Write P for the cone of affine functions nonnegative on $\mathcal{Q}_{222}$. A Bell bound $\beta(q) \leq b$ gives such a function, $f(q) = b - \beta(q)$. We call f a Bell slack. Over a real closed extension K, the transferred cone P_K consists exactly of the affine functions nonnegative on $(\mathcal{Q}_{222})_K$. This follows by transferring the finite formula $\forall q\, [q \in \mathcal{Q}_{222} \Rightarrow f(q) \geq 0]$.

To keep the inequality and the experiment independent, choose two labelled real closed extensions K_c, K_t of $\mathbb{R}$ and put $T = K_c \otimes_{\mathbb{R}} K_t$. For $f = f_0 + \sum_\nu f_\nu q_\nu$ with coefficients in K_c, and $q \in (\mathcal{Q}_{222})_{K_t}$, define

$$[f, q] = f_0 \otimes 1 + \sum_{\nu=1}^{8} f_\nu \otimes q_\nu. \quad (9)$$

An element of T is a finite sum of separated products. We seek an identity $[f, q] = \sum_k h_k^2$ in this ring. The fields may be labelled copies of the same field, but their elements retain their separate positions in the tensor product.

Theorem 2 (Universal sum of squares)**.** *For every $K_c, K_t \supset \mathbb{R}$, every $f \in P_{K_c}$ and every $q \in (\mathcal{Q}_{222})_{K_t}$,*

$$[f, q] \in \Sigma T^2. \quad (10)$$

Here ΣT^2 is the set of finite sums of squares in T.

Scheiderer's Corollary 3.19 [5] states that a closed convex semialgebraic set $S \subset \mathbb{R}^n$ has a finite semidefinite lift precisely when the corresponding tensor evaluations are sums of squares for every real closed extension $K/\mathbb{R}$, every valid affine function over K, and every point of S_K.

Applying Theorem 2 to two labelled copies of each K gives this condition for $\mathcal{Q}_{222}$. Together with Eq. (8), it proves the real-coefficient part of Theorem 1.

The criterion permits the number and tensor ranks of the squares to depend on the point and the affine function. Its conclusion supplies a common finite lift. It is therefore enough to construct exact certificates throughout the source and testing domains. The next two sections make those domains and their certificates explicit.

5 Two families of supporting inequalities

We first reduce the cone of Bell slacks to a small collection of coefficient patterns. Choose the CHSH expression $S = E_{00} - E_{01} + E_{10} + E_{11}$ and set $H = 1 - S/2$. At the corresponding Popescu–Rohrlich vertex, $H = -1$ and the eight losing-event probabilities vanish. Order these events as

$$\begin{array}{c|cccccccc} \text{weight} & \lambda & n & m & C & A & D & \alpha & \beta \\ \text{event} & (++|01) & (+-|00) & (+-|11) & (-+|10) & (-+|00) & (-+|11) & (--|01) & (+-|10) \end{array} \tag{11}$$

The letters on the left denote nonnegative weights, not probabilities. Every valid slack negative at that vertex has the unique form

$$f = dH + 4\sum_i c_i p_i, \qquad d > 0, \quad c_i \geq 0. \tag{12}$$

The coefficients follow by evaluating f at that vertex and its eight deterministic CHSH-facet neighbors. If f is nonnegative at every PR vertex, it is nonnegative on the nonsignalling polytope and is a nonnegative combination of event probabilities. Those slacks already have separated square certificates.

For the remaining slacks, introduce half-angles u, v and a real homogeneous state (W, X, Y, Z). The measurement circle coordinates are $(1-u^2, 2u)/(1+u^2)$ and $(1-v^2, 2v)/(1+v^2)$. Multiplication by the positive measurement denominators and state norm gives the complete polynomial

$$\begin{aligned} F = {} & 4\lambda D_u(W + vX)^2 + 4nD_uD_vX^2 + 4m(vW - X + uvY - uZ)^2 \\ & + 4CD_v(uW - Y)^2 + 4AD_uD_vY^2 + (4D + 2d)(uW + uvX - Y - vZ)^2 \\ & + (4\alpha + 2d)D_u(vY - Z)^2 + (4\beta + 2d)D_v(X + uZ)^2 - 2dD_uD_vZ^2, \end{aligned} \tag{13}$$

where $D_u = 1 + u^2$ and $D_v = 1 + v^2$. Thus $F = D_uD_v\|\psi\|^2 f(\Phi)$ in one field, and the same identity holds with separated coefficients in T. Homogeneous angle pairs give the polynomial at the missing projective endpoints.

The negative term in Eq. (13) is the only obstruction to an immediate event-square decomposition. Let $P(u, v)$ be its four-by-four state matrix. For $d > 0$ and $uv \neq 0$, the restriction to $Z = 0$ contains the positive definite form

$$2d\big[(uW + uvX - Y)^2 + D_uv^2Y^2 + D_vX^2\big]. \tag{14}$$

Hence $P \succeq 0$ is equivalent to $\det P \geq 0$. Its determinant has a useful structure:

$$\det P = 16D_u^2D_v^2u^2v^2L(u^2, v^2), \qquad L(U, V) = h + \sum_{(i,j)\neq(1,1)} q_{ij}U^{i-1}V^{j-1}, \quad q_{ij} \geq 0, \tag{15}$$

where $(i, j) \in \{0, 1, 2\}^2 \setminus \{(1, 1)\}$. The eight coefficients are displayed in Complete proofs, Chapter 4, Eq. (SG6). Their nonnegativity makes $L(e^s, e^t)$ convex. Consequently, a stationary

zero at finite positive angles is a global minimum. Together with Eq. (14), this proves that the six contact equations

$$P(u_c, v_c)\psi_c = 0, \qquad \psi_c^T P_u(u_c, v_c)\psi_c = \psi_c^T P_v(u_c, v_c)\psi_c = 0 \tag{16}$$

already certify physicality of a nonnegative coefficient tuple.

Include the event-face boundary $d = 0$ and normalize all nine nonnegative coefficients in Eq. (12) to sum to one. The physical tuples form a compact convex body. At an extreme point with $d > 0$ whose Laurent polynomial is coercive, the minimum of L must be zero; otherwise small opposite coefficient perturbations would remain physical. At this contact, Eq. (16) gives six linear equations in the source coefficients. Seven positive event weights would leave a nonzero perturbation preserving those equations and the normalization, again contradicting extremality. The compatibility of the omitted events then leaves exactly two patterns, up to Bell relabellings:

$$\begin{aligned} \mathsf{C}_1 &: \quad \lambda = n = 0, \quad d, m, C, A, \alpha > 0, \quad D, \beta \geq 0, \quad D + \beta > 0, \\ \mathsf{C}_2 &: \quad \lambda = \beta = 0, \quad d, m, C, A, D, \alpha > 0, \quad n \geq 0. \end{aligned} \tag{17}$$

The noncoercive supports reduce to two other patterns, $\lambda = m = 0$ or $\lambda = \alpha = 0$. Their source cones have explicit matrix lifts: a six-by-six Gram matrix with three auxiliaries in the first case, and two three-by-three blocks with one auxiliary in the second. Complete proofs, Chapter 4, derives both lifts and their attained dual representations. Pairing a source lift with its testing dual gives separated squares directly.

Every normalized source is a mixture of at most nine extreme sources. This finite decomposition, the contact equations and the support classification all transfer to the coefficient field. It remains to certify the full polynomial (13) for the two families in Eq. (17) at arbitrary independent tests.

6 Exact certificates at regular and singular contacts

6.1 A positive real Gram matrix controls infinitesimal errors

For $K_i \supset \mathbb{R}$, let B_i be its ring of bounded elements and $\mathfrak{m}_i$ its ideal of infinitesimals. Every bounded element has a unique real standard part. This defines a residue map $\rho : B_c \otimes B_t \to \mathbb{R}$ with kernel $\mathfrak{m}_c \otimes B_t + B_c \otimes \mathfrak{m}_t$. Thus a zero-residue coefficient is a finite sum of separated terms, each with an infinitesimal factor.

The following elementary identity turns a positive residue into exact squares. If $x \in K_c$ is infinitesimal, $y \in K_t$ is bounded, and ϵ, M are positive reals with $M > |y|$, then

$$\epsilon + x \otimes y = \tfrac{1}{2}\big[(\epsilon/M + x) \otimes (M + y) + (\epsilon/M - x) \otimes (M - y)\big]. \tag{18}$$

Each pure-field factor on the right is positive, so each product is an elementary tensor square. A positive real residue can be split among finitely many such terms.

Lemma 3 (Bounded Gram matrix)**.** *If $G \in \mathrm{Sym}_r(B_c \otimes B_t)$ and $\rho(G) \succ 0$, then $z^T G z \in \Sigma T^2$ for every $z \in T^r$.*

Proof. A fixed real congruence reduces the residue to the identity. Choose $\delta > 0$ with $(r-1)\delta < 1$. The exact decomposition

$$\begin{aligned} G = &\sum_i (G_{ii} - (r-1)\delta) e_i e_i^T \\ &+ \sum_{i<j} \frac{\delta + G_{ij}}{2}(e_i + e_j)(e_i + e_j)^T + \sum_{i<j} \frac{\delta - G_{ij}}{2}(e_i - e_j)(e_i - e_j)^T \end{aligned} \tag{19}$$

has coefficients of positive residue. Apply Eq. (18) to each coefficient and evaluate on z. The roots in z may be unbounded; only the Gram coefficients require boundedness. □

6.2 Contacts with noncommuting measurement residues

Suppose both local measurement pairs remain noncommuting after taking real residues. At a zero of the Bell slack the residue state must be entangled, since a product state gives a strict CHSH bound at these angles. An entangled two-qubit state is an invertible bilinear form on the local rays. Each local ray can therefore be orthogonal to at most one remote ray. The vanishing losing events form a matching in the cycle of eight events, so at most four of them can vanish.

This finite constraint has an important consequence for unequal source scales. Retain the event amplitudes with the largest coefficients as exact state coordinates. If the next source scale has a positive constant term, Lemma 3 applies. If the signed coefficient is visible at that scale, the constrained contact has a positive Hessian. The angular part follows from Eq. (15): at a stationary zero,

$$\nabla^2_{\log U,\log V} L = \sum_e q_e U^{e_1} V^{e_2}\, e e^T. \tag{20}$$

The active exponent vectors span the plane. The event supports that would reduce their span have an explicit negative CHSH-core direction and are excluded by physicality. The positive transverse state block then gives the full constrained Hessian.

To lift this curvature through independent infinitesimals, write the complete local polynomial as

$$F(x,y) = \sum_i a_i b_i(y) x_i^2 + \sigma G(x,y), \qquad \varepsilon_i = \sigma/a_i > 0 \text{ infinitesimal.} \tag{21}$$

The ratios ε_i may differ arbitrarily. Substituting $x_i = \varepsilon_i s_i$ into the stationary equations gives a nonsingular real residue Jacobian. The semialgebraic implicit-function theorem supplies an exact center (x_c, y_c) in the coefficient field. Finite Taylor expansion there gives

$$F/\sigma = a + h^T Q h, \qquad a \geq 0, \quad \rho(Q) \succ 0, \qquad h = \left((x_i - x_{ci})/\sqrt{\varepsilon_i}\right)_i \oplus (y - y_c). \tag{22}$$

The vanishing first derivatives remove linear terms. Every higher coefficient acquires a bounded or infinitesimal pure-field factor after the displayed rescaling. Lemma 3 gives exact squares of the entire polynomial.

If the signed coefficient remains infinitesimal at the next scale, another positive event amplitude must vanish. Add it to the retained constraints. Each repetition adds an edge to the matching, so the process stops after at most four edges. The four-edge case has an explicit weighted completion and a strictly positive remaining tangential curvature. Complete proofs, Chapter 6, proves this terminal case for arbitrary ratios of the four coefficients.

6.3 Limits of compatible measurements

When a measurement pair becomes compatible in residue, the curvature in Eq. (20) may vanish. The exact contact equations then determine the rescaling needed to keep the lost directions. We first solve those equations for the source coefficients and prove the strict signs of their denominators. The derivation uses a connected graph of real contacts, explicit nonzero minors and event-deletion arguments. It also includes the permitted zero coefficients in Eq. (17); see Complete proofs, Chapter 5.

These coordinates divide the source domain into finitely many cases. The first division uses the signed coefficient relative to the sum of all coefficients. A positive residue gives the ordinary edge and corner cases. An infinitesimal residue gives the two families $\mathsf{C}_1, \mathsf{C}_2$ with further exact source inequalities. Every positive quantity is either infinitesimal, a bounded unit or unbounded in its own field. Literal zero coefficients are kept as separate boundaries. This classification exhausts the source domain without comparing the rates in the two fields.

For each case, choose state deviations ξ from the relevant contact graph and retain the identity

$$F = F_0 + 2\sum_i \xi_i F_i + \sum_{i,j} \xi_i \xi_j F_{ij}. \tag{23}$$

The scalar F_0, all first-normal coefficients F_i and the whole quadratic state form are used together. Contact identities express the vanishing parts through finitely many polynomial roots. Monic division preserves the required source factors; it never divides by a difference between source and testing parameters. Rescaling these roots gives bounded Gram coefficients with a strict real residue. Complementary state charts and measurement directions are treated directly from Eq. (13).

One identity illustrates how the source factors survive this division. Let k be a positive source infinitesimal and s a positive source unit. For independent positive testing coordinates u, v, put

$$U_1 = u^2 v, \qquad U_2 = kuv^2/s, \qquad U_3 = k^2 s, \qquad g = kuv. \tag{24}$$

Then $U_1 U_2 U_3 = g^3$, and the exact identities

$$g(u-k) = k(U_1 - g), \qquad g(v-s) = s(U_2 - g) \tag{25}$$

replace contact differences by the roots already present in the three-monomial sum-of-squares circuit. The first replacement supplies the infinitesimal factor k. In the second, s is a unit, so the corresponding Taylor coefficients must themselves be infinitesimal; the local source formulas establish this before the identity is used. The first contact derivatives cancel the remaining linear terms. Neither $u - k$ nor $v - s$ is inverted. Complete proofs, Appendix B, gives the resulting finite Gram correction and the full state-normal extension.

The resulting statement is the local input needed for the global proof.

Proposition 4 (Complete canonical certificate)**.** *Let $K_c, K_t \supset \mathbb{R}$ be labelled real closed fields. Let the coefficients of Eq.* (13) *belong to either family in Eq.* (17) *and admit a contact satisfying Eq.* (16) *with $u_c, v_c > 0$ and $\psi_c \neq 0$. Then its full homogeneous tensor evaluation is a finite sum of squares for every measurement pair and every state in K_t.*

Chapter 7 of Complete proofs assembles the local certificates. Appendix A treats bounded source coefficients; Appendices B and C establish the two remaining contact families, including their complementary testing charts. Several singular cases share one quadratic completion. In C_1, one contact-division argument allows the source and testing coordinates to have either zero or nonzero residue. Its coefficients retain an infinitesimal factor after every division. In C_2, the four vanishing event gaps enter through one radial scale; an exact current relation cancels the cross term before that scale is divided out. The remaining source domains have the explicit positive matrix completions given there. Source and testing perturbations remain independent throughout.

7 Global assembly and algebraic coefficients

We can now prove Theorem 2. Decompose a valid affine function into finitely many event, closed-face and canonical contact sources. Event sources are separated squares. The explicit closed-face lifts give positive-semidefinite matrices X_c and Y_t over their respective fields. Their pairing is a sum of squares because

$$\sum_{i,j} (X_c)_{ij} \otimes (Y_t)_{ij} = \sum_{a,b} \left(\sum_i U_{ia} \otimes V_{ib} \right)^2, \qquad X_c = UU^T, \quad Y_t = VV^T. \tag{26}$$

The factorizations exist over real closed fields, including for singular matrices. Proposition 4 handles the remaining source families. Dividing out the measurement denominators and the state norm multiplies each root only by positive inverse square roots in the testing field. Source normalizations are restored in the coefficient field.

For completeness, signs and endpoints do not introduce further cases. Local changes of basis give positive finite half-angles. If one pair commutes literally, the CHSH operator is ± 2 times a reflection on each local block, so $H \succeq 0$ in the testing field. At positive half-angles the CHSH core is also nonnegative when $WZ \leq 0$, by its explicit two-state Schur form. The remaining states have $W/Z > 0$ and are covered by Proposition 4. Homogeneous angle coordinates retain both projective endpoints.

Write $f = \sum_j a_j f_j$ with $a_j \geq 0$ in K_c, and use Eq. (8) to write $q = \sum_i w_i q_i$ in K_t. For each source and testing atom, let $[f_j, q_i] = \sum_k h_{ijk}^2$. Then

$$[f, q] = \sum_{i,j,k} \left[(\sqrt{a_j} \otimes \sqrt{w_i}) h_{ijk} \right]^2. \tag{27}$$

The testing weights sum to one and therefore preserve the affine constant. Zero weights give zero roots. This proves Theorem 2, and the tensor criterion proves Eq. (4).

It remains to justify the algebraic entries asserted in Theorem 1. Let $\mathbb{R}_{\mathrm{alg}}$ be the real algebraic numbers. This is a real closed subfield of $\mathbb{R}$. Fix the finite sizes N, p supplied by the existence theorem. Equality between Eq. (8) and a projected pencil of these sizes is a first-order sentence over $\mathbb{Q}$:

$$\exists L_0, \ldots, L_8, M_1, \ldots, M_p \quad \forall q \quad \left[q \in \mathcal{Q}_{222} \iff \exists y \; L_0 + \sum_\nu q_\nu L_\nu + \sum_j y_j M_j \succeq 0 \right]. \tag{28}$$

Here membership uses the finite polynomial formula (8), and matrix positivity uses finitely many principal minors. The sentence is true over $\mathbb{R}$ and hence over $\mathbb{R}_{\mathrm{alg}}$. Choose its matrix coefficients there. Elementary transfer back to $\mathbb{R}$ gives Eq. (4) for every real q, with those same algebraic entries. This completes the proof of Theorem 1.

8 An explicit projection

A two-dimensional projection illustrates what an evaluated representation looks like. Set $t = a_0$ and $s = E_{00} + E_{01} + E_{10} - E_{11}$. The exact projection is

$$\mathcal{T} = \{(t, s) : |t| \leq 1, \quad t^2 + s^2/4 \leq 2\}. \tag{29}$$

It has the order-four linear matrix inequality

$$\begin{pmatrix} \sqrt{2} + t & s/2 \\ s/2 & \sqrt{2} - t \end{pmatrix} \oplus (1 - t) \oplus (1 + t) \succeq 0. \tag{30}$$

The two-by-two determinant gives the ellipse, and the scalar blocks give the interval for t.

The quantum upper boundary follows from the tilted-CHSH identity of Bamps and Pironio [7]. If $T_\alpha = \alpha A_0 + A_0(B_0 + B_1) + A_1(B_0 - B_1)$ and $R_\alpha = \alpha A_1 - A_0(B_0 - B_1) - A_1(B_0 + B_1)$, then

$$T_\alpha^2 + R_\alpha^2 = (8 + 2\alpha^2) I, \qquad \beta I - T_\alpha = \frac{(\beta I - T_\alpha)^2 + R_\alpha^2}{2\beta}, \quad \beta = \sqrt{8 + 2\alpha^2}. \tag{31}$$

For a given $|t| \leq 1$, choosing $\alpha = 2t/\sqrt{2 - t^2}$ gives $|s| \leq 2\sqrt{2 - t^2}$. Conversely, write $r = \sqrt{1 - t^2}$ and $h = \sqrt{2 - t^2}$. The state and measurements

$$|\psi_t\rangle = \sqrt{\frac{1 + t}{2}} |00\rangle + \sqrt{\frac{1 - t}{2}} |11\rangle, \quad A_0 = Z, \quad A_1 = X, \quad B_{0,1} = (Z \pm rX)/h \tag{32}$$

give $(t, s) = (t, 2h)$. Reversing both Bob outcomes gives $(t, -2h)$; mixing these experiments fills the vertical interval. This proves both inclusions in Eq. (29), including $t = \pm 1$. Theorem 1 extends finite semidefinite representability to all eight coordinates at once.

9 Discussion

The complete quantum set for two binary measurements per party has a finite semidefinite representation with real algebraic coefficients. The result places all Bell objectives in this scenario on one exact finite optimization domain. It includes biased outcomes, mixed states and arbitrary finite-dimensional implementations.

The proof depends on retaining the singular limits of the measurement geometry. The real-qubit reduction makes the set semialgebraic; the separated sum-of-squares construction supplies the additional positivity that a finite lift requires. Independent coefficient and measurement parameters expose the cases in which ordinary contact curvature degenerates. Exact rescalings recover those directions while preserving the complete state polynomial.

Finding a small explicit pencil is the next concrete problem. The algebraic-coefficient conclusion makes exact coefficient search possible in principle, while the local identities identify the functions from which a constructive representation may be built. The existence theorem itself is independent of the choice of a moment hierarchy. Its conclusion is that the full quantum boundary in the simplest Bell experiment admits one finite semidefinite description.

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